MIL-C-24643/28D
22 November 1994
SUPERSEDING
MIL-C-24643/28C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1S50MU (INCLUDING VARIATION LS1S50MA AND LS1S50MUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in three variations: LS1S50MU (unarmored), LS1S50MA (armored) and LS1S50MUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, ASTM B 286, size 22-7. maximum diameter 0.033 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Crosslinked polyethylene insulation, white. Nominal wall thickness 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth - Braided shield of AWG no. 36 tin coated copper. Braid angle of 20 to 25 degrees; minimum coverage of 90 percent.
Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.
Sixth - The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch
Seventh - Binder tape, applied helically with overlap.

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Construction (non-watertight) (continued)

Construction variant for type LS1S50MU

Eighth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

Construction variant for type LS1S50MA

Eighth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.
Ninth - Braided metal armor.

Construction variant for type LS1S50MUS

Eighth - Braided shield of tin coated copper. Minimum coverage of 85 percent.
Ninth - Spirally wrapped 0.002 inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.
Tenth - Braided shield of tin coated copper. Minimum coverage of 85 percent.
Eleventh - Binder tape applied helically with overlap.
Twelfth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

TABLE I. Details.

Military 1/ part no. M24643/28	Size 2/	Number of conductors	Diameter over shielded core (in.nom)	Overall diameter (inches)						Cold bending mandrel diameter maximum (inches)
				Type LS1S50MU		Type LS1S50MA		Type LS1S50MUS		
				min	max	min	max	min	max	
-01UN	16	16	0.655	0.760	0.825	0.810	0.875	0.820	0.885	11
-02UN	20	20	.760	.835	.905	.885	.955	.895	.965	12
-03UN	40	40	1.015	1.095	1.185	1.145	1.235	1.155	1.245	15
-04UN	70	70	1.380	1.465	1.555	1.515	1.605	1.525	1.615	20

1/ Unarmored, overall shield. When armor option or double shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LS1S50MU, LS1S50MA and LS1S50MUS.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum.....	17.54
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	5000
Shield to shield (component shields only)	500
Shield to shield (type LS1S50MUS overall shields only)	200

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EXAMINATION AND TESTS (contineud):	Requirements
Insulation resistance - Megohms/1000 feet minimum.....	
Conductor to shield	2500
shield to shield (component)	100
shield to shield (LS1S50MUS only, overall shields)	100
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Characteristic impedance - at 1MHz, ohms	50 ± 2
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C).....	50
Attenuation - at 400 Mhz, decibel (db)/100 FT maximum	18
Cold bending, cable - at -20 ± 2°C, (see table I for mandrel diameter)	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles, minimum	125
Armor (type LS1S50MA only) - conformance to material, construction and coverage	No failure

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EXAMINATION AND TESTS (contineud): Requirements

Group C (continued):

Shield - conformance to material, construction and coverage	No failure
Surface transfer impedance (type LS1S50MUS only)	
Milliohms per meter, maximum	700
Emp response time - db, minimum	60

Group D:

Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project No. 6145-2101-28)

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